SUPERSTORE SALES ANALYSIS EXCEL PROJECT

Business Problem

In today's competitive business environment, organizations generate large volumes of sales data, but raw data alone provides little value without proper analysis. Businesses need to understand sales performance, customer behavior, product profitability, and regional trends to make informed decisions that improve profitability and operational efficiency.

This project demonstrates how Microsoft Excel can be used to transform raw sales data into meaningful business insights. Through data preparation, exploration, analysis, and interactive dashboards, the project identifies key trends and provides data-driven recommendations to support strategic business decision-making.

Data Source

Kaggle: https://www.kaggle.com/datasets/ishanshrivastava28/superstore-sales

Business Questions

1. Which states and regions generate the highest sales and profit?
2. How do sales and profit change over time?
3. Which product categories and sub-categories contribute the most profit?
4. Which customer segments contribute the most profit?
5. Does offering larger discounts reduce profitability?

Data Preparation

Data Cleaning

- Removed duplicates (there were none).
- Ensured there are no missing values by using COUNTBLANK formula.
- Trimmed all text/categorical columns by using TRIM formula.
- Ensured all date columns (order date, ship date) are in short date data type. (changed from General --> Short Date).

- Changed sales, profit and discount to Number data type (General --> Number). Justification is that other systems might not recognize data types in Excel like Percentage or Currency.
- Resolved date columns (order date, ship date) data type issue by using formula (example):

=IF(ISNUMBER(C4),C4,DATE(RIGHT(C4,4),MID(C4,4,2),LEFT(C4,2)))
,then change to Short Date data type.

Data Transformation

- Created profit margin column by diving profit with sales, and changed its data type to Number (General --> Number).
- Created Order Year column from Order Date column by using YEAR() formula.
- Created Order Month column from Order Date column by using TEXT() formula. (e.g. TEXT("C4","mmm")
- Created Order Quarter column from Order Date column by using the following formula (example):
  "Q"&ROUNDUP(MONTH(C2)/3,0)

Data Enrichment

- Created a new sheet named "regional sales manager", and inserted sales manager names for each region, then used XLOOKUP formula to insert sales manager column in the main sheet (formula used shown as example below):
  XLOOKUP(O2,'regional sales manager'!A$2:A$5,'regional sales manager'!B$2:B$5)

Exploratory Data Analysis

Business Insights

1. What do the overall sales and profit performance look like?

Total sales amounted to approximately $2.3 million, with an average sales value of $229.86 per transaction. Sales ranged from as low as $0.44 to as high as $22.6 thousand, indicating a wide variation in transaction values. This suggests that customers made both low-value and high-value purchases, reflecting a diverse range of purchasing behaviors.

The business generated a total profit of approximately $286 thousand, with an average profit of $28.66 per order. The lowest-profit transaction incurred a loss of approximately

$6.6 thousand, which, upon further investigation, was associated with a product sold at a 70% discount. In contrast, the highest-profit transaction generated a profit of approximately $8.4 thousand. These figures indicate a significant variation in transaction profitability, suggesting that factors such as discounting may have a substantial impact on business performance.

2. Which shipping modes are used the most?

Standard Class was the most frequently used shipping method, accounting for 5,968 orders, followed by Second Class (1,945 orders) and First Class (1,538 orders). Same Day delivery was the least utilized, with only 543 orders. These results suggest that customers generally preferred lower-cost or standard shipping options over expedited delivery, which may help management optimize shipping operations and service offerings.

3. Which categories/subcategories of products generated loss?

Among all product sub-categories, Tables, Bookcases, and Supplies were the only ones to generate an overall loss, with approximately $17.7 thousand, $3.5 thousand, and $1.2 thousand, respectively. A preliminary review indicates that these sub-categories frequently received discounts. Therefore, the relationship between discount rates and profitability will be examined further in the final dashboard to determine whether discounting contributed to these losses.

4. Who is the top performing sales manager?

Alyss generated the highest total sales, contributing approximately $725 thousand, and was responsible for the West region. In contrast, Rachel recorded the lowest total sales at approximately $392 thousand. These results provide management with a benchmark for comparing sales performance across regions and identifying opportunities to improve sales strategies and resource allocation.

5. Which product categories/subcategories are sold the most?

Within the Furniture category, Furnishings and Chairs recorded the highest quantities sold, with over 3.5 thousand and 2.3 thousand units, respectively. For Office Supplies, Binders and Paper were the best-selling sub-categories, with approximately 6.0 thousand and 5.2 thousand units sold. In the Technology category, Phones and Accessories led sales, with approximately 3.3 thousand and 3.0 thousand units sold, respectively. These findings help management identify high-demand products and support inventory planning to ensure sufficient stock availability while informing future product and procurement strategies.

Summary of EDA

Exploratory Data Analysis (EDA) was conducted using Tables to explore the dataset and identify key business patterns prior to developing the final dashboard. The analysis examined overall sales and profit performance, shipping method preferences, loss-making product sub-categories, sales performance by sales manager, and product demand based on quantities sold.

The EDA revealed several meaningful insights, including substantial variation in transaction values and profitability, a strong customer preference for Standard Class shipping, a small number of product sub-categories operating at a loss, differences in sales performance across regions and sales managers, and consistently high demand for selected product sub-categories. These findings provided an initial understanding of the business and helped identify areas for deeper analysis, which are further explored through interactive visualizations and recommendations in the final dashboard.

Data Analysis (via Dashboards)

1. Which states and regions generate the highest sales and profit?

The West is the strongest-performing region, generating both the highest sales and the highest profit, followed by the East. At the state level, California leads both sales and profit within the West, while New York performs similarly in the East.

However, high sales do not always translate into high profit. In the Central region, Texas records the highest sales but operates at a loss, suggesting that high revenue alone is not sufficient to ensure profitability. Similarly, in the South, Florida generates the highest sales, yet Virginia achieves higher profit despite lower revenue, indicating more efficient cost management or a more profitable sales mix.

Overall, the results suggest that profitability should be evaluated alongside sales, as regions or states with strong revenue may still underperform financially.

2. How do sales and profit change over time?

Both sales and profit fluctuate from quarter to quarter but exhibit a clear upward trend over the four-year period. Sales increased from approximately 96K in 2011 Q1 to 219K in 2014 Q4, while profit grew from 7K to 27K over the same period. This indicates that the business has experienced sustained growth in both revenue and profitability despite short-term fluctuations.
While overall performance is positive, fluctuations in profitability suggest that growth is influenced by underlying business factors. To better understand what drives profitability, the next step is to analyze product performance, customer segments, and the impact of discounts.

3. Which product categories and sub-categories contribute the most profit?

Technology is the most profitable product category, driven primarily by Copiers, while Furniture generates the lowest profit despite containing high-performing products such as Chairs. Within Furniture, some sub-categories generate loss, indicating inconsistent profitability across the category. Management should prioritize investment in consistently high-profit products like Copiers while reviewing underperforming sub-categories to improve pricing strategies, reduce costs, or reconsider the product mix.

4. Which customer segments contribute the most profit?

Consumers generate the highest total profit, followed by Corporate and Home Office customers. However, this ranking closely mirrors the distribution of customer segments, with Consumers accounting for the largest share of orders, followed by Corporate and Home Office, as shown in the pie chart. This suggests that the higher total profit from the Consumer segment is primarily driven by a higher volume of orders rather than greater profitability on a per-order basis. To compare the inherent profitability of each segment more fairly, additional metrics such as average profit per order or profit margin should be considered.

5. Does offering larger discounts reduce profitability?

Average profit peaks at a 10% discount and generally declines as discount levels increase. Beyond approximately 22%, orders become unprofitable on average, with the largest average loss of $310.70 occurring at a 50% discount. This indicates a strong negative relationship between discount levels and profitability, suggesting that aggressive discounting significantly erodes profit margins. Management should carefully evaluate discounts above 20% and ensure they are supported by strategic objectives, such as clearing excess inventory or driving customer acquisition, rather than applying them routinely.

Key Insights

1. Which states and regions generate the highest sales and profit?

The West is the top-performing region, led by California, while the East ranks second, driven by New York. High sales do not always result in high profit, as seen in Texas, which records high sales but operates at a loss.

2. How do sales and profit change over time?

Sales and profit show a clear upward trend from 2011 to 2014, despite quarterly fluctuations. This indicates sustained business growth in both revenue and profitability.

3. Which product categories and sub-categories contribute the most profit?

Technology, led by Copiers, generates the highest profit, while Furniture is the least profitable due to loss-making sub-categories. This highlights opportunities to optimize underperforming products.

4. Which customer segments contribute the most profit?

The Consumer segment generates the highest total profit, followed by Corporate and Home Office. However, this largely reflects a higher volume of orders rather than inherently higher profitability per order.

5. Does offering larger discounts reduce profitability?

Average profit declines as discount levels increase, with orders becoming unprofitable beyond approximately 22%. This suggests that excessive discounting significantly reduces profitability and should be applied selectively.

Recommendations

Superstore should prioritize investment in high-performing products and regions while addressing underperforming areas. Technology, particularly Copiers, consistently delivers strong profitability, whereas loss-making sub-categories such as Tables and Bookcases should be reviewed to optimize pricing, reduce costs, or refine the product mix. Additionally, successful strategies from high-performing regions such as the West could be evaluated for adoption in lower-performing regions.

The company should adopt a more disciplined discount strategy to protect profitability. Since average profit declines significantly beyond approximately 20% discounts, higher discount levels should be reserved for specific objectives such as inventory clearance or customer acquisition. Additionally, customer segments should be evaluated using profit-related metrics, such as average profit per order and profit margin, to determine whether differences in profitability are driven by customer behavior or simply by higher order volumes.

Skills Demonstrated

Formulas, XLOOKUP, Pivot Tables, Conditional Formatting, Charts